

Video frame rate can vary from five to 30 frames per second (fps), depending on the quality of the Internet connection, the hardware, and the software used. ISDN transmits data at minimum rates of 128 Kbps per line and ensures dedicated bandwidth for smooth audio and video (15 to 30 fps).

IP-based video-conferencing solutions rely on the nature and quality of bandwidth provided by the ISP. Lags or delays that cause jerky videos or substandard audio quality are a result of the poor quality of Internet connections. Although Web-based conferencing solutions are expected to eventually replace ISDN as the medium of choice for video conferencing, ISDN is still widely recommended for its high quality.

Voice, Data And More

Apart from conventional video-conferencing solutions, most vendors offer Web conferencing solutions—Microsoft's Net Meeting software, for example, is a PC-based Web conferencing solution; audio conferencing solutions using a Public Switched Telephone Network (PSTN) or Voice over IP (VoIP), IP video conferencing (over ISDN), and streaming video can be used to create interactive presentations from your boardroom.

Apart from just the ability to communicate audio-visually, some vendors also provide

customised tools for document transfer, creation of presentations and incorporation of various media such as AV clips, graphics, animations, etc, in real-time, during an AV session.

Good For Your Company?

Video-conferencing has made age-old corporate boardroom meetings a thing of the past. Access to requisite bandwidth is now easier than ever in India, at least for corporates.

Yugal Sharma, country manager, Polycom India, a company that facilitates video-conferencing, asserts that video-conferencing is one of the best solutions for remote meetings. Says Sharma, "Video-conferencing provides an excellent tool to improve communications with multiple sites within a business, or with other businesses. Also, in applications such as meetings, training, or services, which can all be enhanced through face-to-face communications, video-conferencing works best."

Businesses that operate in various nations also derive huge benefits from video-conferencing. "In the service sector, for instance, video-conferencing can enhance customer experience, improve operational efficiency and create significant savings in cost incurred on travel and time," says Sharma.

Noida-based Havell's India



Video conferencing has enabled us to expand our market

Vivek Khanna
General Manager,
Havell's India

Ltd. is a multi-product electric engineering organisation specializing in solutions for low-voltage electrical requirements. Havell's India has offices across the country and uses video-conferencing solutions in twenty-four locations.

Says Vivek Khanna, general manager (IT and Finance) at Havell's, "We use video-conferencing for inter office communication, and for our marketing, financial and sales needs. Video-conferencing has enabled us expand our market. Employees in branch offices are also able to interact with dealers and can connect with senior management or product heads anytime, from anywhere."

Video-conferencing, in this case, serves as a tool to facilitate better communication with remote customers, partners and offices, thus enabling efficient business transactions.

On the flip side, although vendors claim to conduct interoperability tests with their competitors (so as to not restrict the users to a particular brand), compatibility often becomes an issue. Both ends—the company and clients—may be forced to use similar equipment, which may not always be affordable.

Due to the exorbitant cost of video-conferencing equipment, some companies such as WebEx and Tata Telecom provide hardware and services to corporate

Setting Up A Video-conference Over An ISDN Line

1 STEP Set up your workstation—a PC with a sound card, a set of speakers, a microphone, a webcam with a video-capture card and relevant software, and a dedicated ISDN line

2 STEP Download/purchase and install a video conferencing software such as Microsoft's LiveMeeting or WebEx Client (for PC and Mac users). Follow instructions to set up the software on your desktop. Ensure that the software on the client's end is compatible with yours

3 STEP Set preferences such as speed, picture quality and sound quality. Select or add the client's ID from the video-conferencing program and dial his/her number. Once both sides are connected, you can share visual data.

To share audio data, you will require a telephone connection, along with an ISDN line for data transfer. But this is a rather tedious procedure and you not have access to a dedicated ISDN line.

Meeting Over A Chat Client

Standard chat software (MSN and Yahoo! Messenger, for example) can be used to conduct a Web conference over your local network or over the Internet. Although both audio and visual data transfer is enabled, the quality of the conference will depend on the bandwidth offered by your ISP.

Server-Based Solutions

Companies can also consider server-based AV transfer solutions such as Macromedia Flash Communication Server MX, Microsoft Live Meeting or



Convoq. These are server-based platforms that enable creation of interactive web-based AV applications like video-conferences, live broadcasts, and real-time collaboration applications. Data transfer is relatively secure as it is not over a public connection and is not a major security concern.